

CBSE 2027 Physics — Predicted Paper (Set 4 of 10)

SECTION A: Objective Type Questions (16 Marks)

This section consists of 12 MCQs and 4 Assertion-Reasoning questions of 1 mark each.

Q1. The torque acting on an electric dipole of dipole moment $\vec{p}$ placed in a uniform electric field $\vec{E}$ is maximum when the angle between $\vec{p}$ and $\vec{E}$ is: (a) 0° (b) 45° (c) 90° (d) 180°
(Chapter: Electric Charges and Fields) | **[Confidence: Very High]**

Q2. The temperature coefficient of resistivity (α) for standard alloy resistors like manganin and nichrome is: (a) very small and positive. (b) very small and negative. (c) very large and positive. (d) very large and negative. (Chapter: Current Electricity) | **[Confidence: High]**

Q3. A proton and an alpha particle enter a uniform magnetic field perpendicularly with the same linear momentum. The ratio of the radii of their circular paths ($r_p : r_\alpha$) will be: (a) $1 : 2$ (b) $2 : 1$ (c) $1 : 1$ (d) $4 : 1$ (Chapter: Moving Charges and Magnetism) | **[Confidence: Very High]**

Q4. The magnetic susceptibility (χ) of a paramagnetic substance is: (a) small and negative. (b) small and positive. (c) large and positive. (d) large and negative. (Chapter: Magnetism and Matter) | **[Confidence: Very High]**

Q5. The soft iron core of a transformer is laminated to minimize energy loss due to: (a) hysteresis. (b) flux leakage. (c) eddy currents. (d) resistance of the primary coil. (Chapter: Electromagnetic Induction) | **[Confidence: Very High]**

Q6. In a plane electromagnetic wave, the ratio of the amplitude of the electric field (E_0) to the amplitude of the magnetic field (B_0) is equal to: (a) Planck's constant (h). (b) the speed of light in vacuum (c). (c) the refractive index of the medium (μ). (d) the intrinsic impedance of free space. (Chapter: Electromagnetic Waves) | **[Confidence: Very High]**

Q7. A convex lens made of glass (refractive index $\mu_g = 1.5$) has a focal length f in air. If it is completely immersed in water (refractive index $\mu_w = 1.33$), its focal length will: (a) decrease. (b) remain the same. (c) increase. (d) become negative (act as a concave lens). (Chapter: Ray Optics and Optical Instruments) | **[Confidence: Very High]**

Q8. In a single-slit diffraction experiment, if the width of the slit is halved, the angular width of the central maximum will: (a) be halved. (b) be doubled. (c) remain unchanged. (d) become four times. (Chapter: Wave Optics) | **[Confidence: Very High]**

Q9. Photons of energy 5.0 eV fall on the surface of a metal whose work function is 3.0 eV . The stopping potential required to stop the fastest photoelectrons is: (a) 2.0 V (b) 8.0 V (c) 1.5 V (d) 5.0 V (Chapter: Dual Nature of Radiation and Matter) | **[Confidence: Very High]**

Q10. Which of the following spectral series of the hydrogen atom lies entirely in the visible region of the electromagnetic spectrum? (a) Lyman series (b) Balmer series (c) Paschen series (d) Brackett series (Chapter: Atoms) | **[Confidence: Very High]**

Q11. According to the nuclear drop model, the radius (R) of a nucleus is related to its mass number (A) as: (a) $R \propto A$ (b) $R \propto A^{1/2}$ (c) $R \propto A^{1/3}$ (d) $R \propto A^2$ (Chapter: Nuclei) |

[Confidence: Very High]

Q12. In a p-type semiconductor, the majority charge carriers and the type of dopants added are respectively: (a) Electrons, Pentavalent (b) Holes, Pentavalent (c) Electrons, Trivalent (d) Holes, Trivalent (Chapter: Semiconductor Electronics) | **[Confidence: Very High]**

Q13. Assertion (A): Electric field lines never form closed continuous loops. **Reason (R):** The electrostatic field is a conservative field, and the work done in moving a charge along a closed path is zero. (Chapter: Electric Charges and Fields) | **[Confidence: Very High]**

Q14. Assertion (A): Magnetic field lines always form continuous closed loops. **Reason (R):** Magnetic monopoles do not exist in nature; every magnet acts as a dipole. (Chapter: Magnetism and Matter) | **[Confidence: Very High]**

Q15. Assertion (A): In an interference pattern, energy is created at the maxima and destroyed at the minima. **Reason (R):** The phenomenon of interference violates the law of conservation of energy. (Chapter: Wave Optics) | **[Confidence: High]**

Q16. Assertion (A): The resistivity of a typical intrinsic semiconductor decreases exponentially with an increase in temperature. **Reason (R):** At higher temperatures, more covalent bonds break, exponentially increasing the number density of free charge carriers. (Chapter: Semiconductor Electronics) | **[Confidence: Very High]**

SECTION B: Very Short Answer Type Questions (10 Marks)

This section consists of 5 questions of 2 marks each.

Q17. Plot a graph showing the variation of Coulomb force (F) versus $(1/r^2)$, where r is the distance between a pair of identical point charges. What does the slope of this graph represent? (Chapter: Electric Charges and Fields) | **[Confidence: High]**

Q18. Calculate the magnitude of the magnetic moment associated with an electron revolving in a circular orbit of radius r with speed v . (Chapter: Moving Charges and Magnetism) | **[Confidence: Very High]**

Q19. A circular coil of N turns and cross-sectional area A is rotated with a constant angular velocity ω in a uniform magnetic field B . Under what orientation is the induced EMF in the coil: (i) maximum, and (ii) zero? (Chapter: Electromagnetic Induction) | **[Confidence: Medium]**

Q20. State any two essential conditions for obtaining a sustained and observable interference pattern of light in Young's Double Slit Experiment. (Chapter: Wave Optics) | **[Confidence: High]**

Q21. If the radius of the ${}^{125}_{53}\text{Te}$ nucleus is 7.5 fm , calculate the radius of the ${}^{27}_{13}\text{Al}$ nucleus using the appropriate empirical relation. (Chapter: Nuclei) | **[Confidence: Very High]**

SECTION C: Short Answer Type Questions (21 Marks)

This section consists of 7 questions of 3 marks each.

Q22. Derive an expression for the potential energy of an electric dipole of dipole moment $\vec{p}$ placed in a uniform external electric field $\vec{E}$. Identify the orientations corresponding to stable and unstable equilibrium. (Chapter: Electrostatic Potential and Capacitance) | **[Confidence: Very High]**

Q23. A battery of EMF 10 V and internal resistance 3Ω is connected to an external resistor. If the steady current in the circuit is 0.5 A , calculate: (i) the resistance of the external resistor, and (ii) the terminal voltage of the battery. (Chapter: Current Electricity) | **[Confidence: Very High]**

Q24. Define the term 'mutual inductance' between two coils. Derive an expression for the mutual inductance of two long, coaxial solenoids of the same length wound one over the other. (Chapter: Electromagnetic Induction) | **[Confidence: Very High]**

Q25. State Ampere's Circuital Law. Use it to find an expression for the magnetic field magnitude deep inside a long straight solenoid carrying a steady current I and having n turns per unit length. (Chapter: Moving Charges and Magnetism) | **[Confidence: Very High]**

Q26. Prove mathematically that the average power dissipated in a purely capacitive AC circuit over a complete cycle is zero. (Chapter: Alternating Current) | **[Confidence: High]**

Q27. Two harmonic waves of monochromatic light represented by $y_1 = a \cos(\omega t)$ and $y_2 = a \cos(\omega t + \phi)$ superimpose on each other. Derive the expression for the resultant intensity. Also, write the condition (in terms of phase difference ϕ) for constructive interference. (Chapter: Wave Optics) | **[Confidence: Very High]**

Q28. Using energy band diagrams, clearly distinguish between an n-type semiconductor and a p-type semiconductor. Indicate the position of the donor and acceptor energy levels in the respective diagrams. (Chapter: Semiconductor Electronics) | **[Confidence: Very High]**

SECTION D: Case-Study Based Questions (8 Marks)

This section consists of 2 questions of 4 marks each. Questions usually have internal sub-parts.

Q29. Case Study 1: The Compound Microscope Passage: A compound microscope is an optical instrument used to see highly magnified images of tiny objects. It consists of two convex lenses of short focal lengths arranged coaxially: the objective lens (closer to the object) and the eyepiece (closer to the eye). The objective forms a real, inverted, and magnified image of the object, which acts as the object for the eyepiece. The eyepiece then functions like a simple magnifier to form the final enlarged virtual image. (a) What is the nature of the intermediate image formed by the objective lens? (1 mark) (b) Why are the objective and eyepiece of a compound microscope chosen to have very short focal lengths? (1 mark) (c) In a compound microscope, the objective has a focal length of 1.25 cm and the eyepiece has a focal length of 5 cm . An object is placed 1.5 cm from the objective. If the final

image is formed at infinity (normal adjustment), calculate the magnifying power of the microscope. (Assume $D = 25 \text{ cm}$). **OR** How does the magnifying power of the microscope change if the focal length of the objective lens is decreased? Justify mathematically. (2 marks) (Chapter: Ray Optics and Optical Instruments) | **[Confidence: Very High]**

Q30. Case Study 2: Photoelectric Effect & Lenard's Observations *Passage:* Lenard and Hallwachs studied the photoelectric effect experimentally. They observed that when ultraviolet light falls on an emitter plate, electrons are ejected. They studied the variation of photoelectric current with the intensity of light, the potential of the anode, and the frequency of incident radiation. They found that for a given photosensitive material, there exists a certain minimum cut-off frequency below which no photoelectrons are emitted, regardless of the light's intensity. (a) How does the saturation photoelectric current vary with the intensity of incident light (assuming frequency is kept constant above threshold)? (1 mark) (b) What is the effect of increasing the frequency of incident light on the stopping potential? (1 mark) (c) Light of wavelength $4000 \text{ \AA}$ is incident on a metal surface of work function 2.0 eV . Calculate the maximum kinetic energy of the emitted photoelectrons. (Take $hc \approx 1240 \text{ eV} \cdot \text{nm}$). **OR** Draw a labelled graph showing the variation of stopping potential with the frequency of incident radiation for two different metals A and B having work functions $W_A > W_B$. (2 marks) (Chapter: Dual Nature of Radiation and Matter) | **[Confidence: Very High]**

SECTION E: Long Answer Type Questions (15 Marks)

This section consists of 3 questions of 5 marks each. All questions have an internal "OR" choice.

Q31. (a) Derive an expression for the equivalent capacitance of three capacitors C_1 , C_2 , and C_3 connected in parallel across a voltage source V . **(b)** A parallel plate capacitor with air between its plates has a capacitance of 8 pF . What will be its new capacitance if the distance between the plates is reduced by half, and the entire space between them is filled with a dielectric material of dielectric constant $K = 6$? **OR (a)** State Gauss's theorem in electrostatics. **(b)** Two large, thin, and parallel metal plates are placed close to each other. On their inner faces, the plates have uniform surface charge densities of opposite signs but equal magnitude $\sigma = 17.7 \times 10^{-22} \text{ C/m}^2$. Calculate the magnitude of the electric field $\vec{E}$: (i) in the outer region of the first plate. (ii) in the space between the two plates. (Chapter: Electrostatics / Electric Charges and Fields) | **[Confidence: Very High]**

Q32. (a) Draw a schematic labelled sketch of a step-up transformer and state its working principle. **(b)** Deduce the mathematical expression for the ratio of the secondary voltage to the primary voltage in terms of the number of turns in the two coils, assuming it to be an ideal transformer. **(c)** Mention any two causes of energy loss in a real, practical transformer and state how they can be minimized. **OR (a)** With the help of a neat labelled diagram, explain the principle and working of an AC Generator. **(b)** A circular coil of cross-sectional area 200 cm^2 and 20 turns is rotated about its vertical diameter with an angular speed of 50 rad/s in a uniform horizontal magnetic field of magnitude $3 \times 10^{-2} \text{ T}$. Calculate the

maximum value of the EMF induced in the coil. What is the average value of the EMF induced over a complete cycle? (*Chapter: Alternating Current / Electromagnetic Induction*) | **[Confidence: Very High]**

Q33. (a) Describe the single-slit diffraction experiment using Huygens' principle. State the conditions (in terms of path difference) for the formation of the principal maximum, secondary minima, and secondary maxima on the screen. **(b)** Derive the expression for the linear width of the central maximum on a screen kept at a distance D from the slit of width a . **(c)** How will the width of the central maximum be affected if the slit width ' a ' is doubled? **OR** **(a)** Draw a neat, labelled ray diagram showing the image formation by an Astronomical Telescope when the final image is formed at infinity (normal adjustment). **(b)** Obtain the expression for its magnifying power in normal adjustment. **(c)** Give two explicit reasons why reflecting telescopes are generally preferred over refracting telescopes in modern astronomy. (*Chapter: Wave Optics / Ray Optics and Optical Instruments*) | **[Confidence: Very High]**

CBSE Class 12 (2027) — AI-Predicted via PYQ/Blueprint Analysis. Probability-weighted guidance, not actual exam questions.